

END SEM QUESTIONS: PART A – HINTS

Question 1. (2 + 1 marks) Let i be an aperiodic, positive recurrent state in a Markov Chain with the mean recurrence time 4. Then, $\boxed{\lim_{n \rightarrow \infty} p_{ii}^n = \frac{1}{4}}$ and $\boxed{\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{m=1}^n p_{ii}^m = \frac{1}{4}}$

Question 1: ALTERNATIVE. (2 + 1 marks) Let i be an aperiodic, positive recurrent state in a Markov Chain with the mean recurrence time 2. Then, $\boxed{\lim_{n \rightarrow \infty} p_{ii}^n = \frac{1}{2}}$ and $\boxed{\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{m=1}^n p_{ii}^m = \frac{1}{2}}$

Question 2. (1 + 2 marks) Let i be a positive recurrent state in a Markov Chain with the mean recurrence time 4. Suppose period of i is 2. Then, $\boxed{\lim_{n \rightarrow \infty} p_{ii}^n \text{ does not exist}}$ and $\boxed{\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{m=1}^n p_{ii}^m = \frac{1}{4}}$

Note: $\lim_{n \rightarrow \infty} p_{ii}^{2n} = \frac{2}{4}, \lim_{n \rightarrow \infty} p_{ii}^{2n+1} = 0$

Question 3. (1 + 1 marks) Let i be a null recurrent state in a Markov Chain with period 3. Then, $\boxed{\lim_{n \rightarrow \infty} p_{ii}^{3n} = 0}$ and $\boxed{\lim_{n \rightarrow \infty} p_{ii}^{3n+1} = 0}$

Question 4. (1 mark) Suppose i and j be two states in a Markov Chain such that i is recurrent and $i \leftrightarrow j$. Then, $\boxed{f_{ij} = 1}$

Question 5. (1 mark) Suppose that a recurrent Markov Chain with the state space $S = \{0, 1, 2, \dots\}$ is known to be irreducible. If $\sum_{n=1}^{\infty} n f_{11}^n = \infty$, then the number of null recurrent classes in this Markov Chain is $\boxed{1}$

Note: There is exactly one communication class. Since the state 1 is null recurrent, the class is null recurrent.

Question 6. $\{X_n\}_{n=0}^{\infty}$ be a Markov Chain on the state space $S = \{0, 1, 2, 3, 4, 5\}$ and with the transition probability matrix

$$P = \begin{pmatrix} \frac{1}{3} & \frac{2}{3} & 0 & 0 & 0 & 0 \\ \frac{2}{3} & \frac{1}{3} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{1}{4} & \frac{3}{4} & 0 & 0 \\ 0 & 0 & \frac{3}{4} & \frac{1}{4} & 0 & 0 \\ \frac{1}{4} & 0 & \frac{1}{4} & 0 & 0 & \frac{1}{2} \\ \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{3} & 0 \end{pmatrix}$$

- (a) (2 marks) The number of communication classes is $\boxed{3}$
- (b) (1 mark) The number of communication classes, which are closed, is $\boxed{2}$
- (c) (2 marks) The number of positive recurrent classes is $\boxed{2}$

Note: communication classes are $\{0, 1\}, \{2, 3\}, \{4, 5\}$. The first two are positive recurrent and hence closed, the last one is transient.

Question 7. Underline ‘Yes’ or ‘No’ as appropriate in the following sub-questions.

- (a) (2 marks) In an aperiodic Markov Chain with finite state space S and transition probability matrix $(p_{ij})_{i,j \in S}$, it is known that there is a transient state. Consider the statement: ‘There exists a proper subset $C \subsetneq S$ such that $(p_{ij})_{i,j \in C}$ is stochastic.’ Is the statement true? $\boxed{\text{Yes/No.}}$

Note: Take C to be any positive recurrent class.

- (b) (2 marks) Consider any Markov Chain $\{X_n\}_{n \geq 0}$ with a countably infinite state space and with exactly one absorbing state and at least one transient state. Let i and j denote the absorbing and a transient state, respectively. Consider the statement: ‘There exists examples of such Markov Chains such that $\mathbb{P}(\{X_n\}_{n \geq 1} \text{ is absorbed into } i \mid X_0 = j) < 1$.’ Is the statement true? $\boxed{\text{Yes/No.}}$

Note: Take the asymmetric random walk on $\{0, \pm 1, \pm 2, \dots\}$ in one dimension – this is transient.

Attach an extra point $\{*\}$ and look at the state space $S = \{*, 0, \pm 1, \pm 2, \dots\}$. Consider the transitions on S as follows:

$$p_{**} = 1,$$

$$p_{j*} = p_{*j} = 0, \forall j \in \{0, \pm 1, \pm 2, \dots\}$$

$$p_{jk} \text{ as in the random walk, } \forall j, k \in \{0, \pm 1, \pm 2, \dots\}$$

Then, $\mathbb{P}(X_n \in \{0, \pm 1, \pm 2, \dots\}, \forall n | X_0 = j) = 1$, the state $*$ is absorbing and

$$\mathbb{P}(\{X_n\}_{n \geq 1} \text{ is absorbed into } * | X_0 = j) = 0.$$

Question 8. In this problem, we consider red and green balls. Balls are identical except for colour and balls of the same colour are identical. Suppose that an urn contains one red ball and two green balls. Consider the following random experiment: a ball is drawn from the urn at random and is replaced with a new ball of the other colour, i.e. a drawn red ball is replaced by a new green ball and vice versa. At the time $n = 0$, we do not perform this experiment and instead look at the initial state as mentioned earlier. At the time points $n = 1, 2, \dots$, the random experiment is performed once. Denote by X_n the number of red balls in the urn at the time $n = 0, 1, 2, \dots$ and consider the Markov Chain $\{X_n : n = 0, 1, 2, \dots\}$.

(a) (1 mark) The state space is $\{0, 1, 2, 3\}$

(b) (2.5 marks) The transition probability matrix is

$$\begin{pmatrix} 0 & 1 & 0 & 0 \\ \frac{1}{3} & 0 & \frac{2}{3} & 0 \\ 0 & \frac{2}{3} & 0 & \frac{1}{3} \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

(c) (1 mark) Period of state 0 is 2

(d) (1 mark) $\mathbb{P}(X_{n+2} = 2 | X_n = 1) = 0$

(e) (1 mark) Is the Markov Chain irreducible? Underline the correct answer. Yes/No.

(f) (2.5 marks) Write down a stationary distribution for this Markov Chain as a row vector $(\frac{1}{8}, \frac{3}{8}, \frac{3}{8}, \frac{1}{8})$

(g) (1 mark) Is the above stationary distribution unique? Underline the correct answer. Yes/No.

Note: the Markov Chain is irreducible and the state space is finite. Hence, it is also positive recurrent. Since, there is a unique positive recurrent class, a stationary distribution exists and is unique. This can be obtained from the system of equations:

$$\pi = \pi P, \sum_{i=0}^3 \pi_i = 1, \pi_i \geq 0, \forall i \in \{0, 1, 2, 3\}.$$

Question 9. Consider a Markov chain $\{X_n\}_{n=0}^{\infty}$ with state space $S = \{0, 1, 2, 3, 4\}$ and the transition probability matrix

$$P = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ \frac{1}{4} & 0 & \frac{3}{4} & 0 & 0 \\ 0 & \frac{1}{4} & 0 & \frac{3}{4} & 0 \\ 0 & 0 & \frac{1}{4} & 0 & \frac{3}{4} \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

(a and b) (1 mark) Period of state 0 is 1, Period of state 2 is 2, Period of state 3 is 2, Period of state 4 is 1.

- (c) (4 marks)
- | |
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| $\mathbb{P}(\{X_n\}_{n \geq 1} \text{ is absorbed into } \{0\} \mid X_0 = 1) = \frac{13}{40}$
ALTERNATIVE: $\mathbb{P}(\{X_n\}_{n \geq 1} \text{ is absorbed into } \{4\} \mid X_0 = 1) = \frac{27}{40}$
ALTERNATIVE: $\mathbb{P}(\{X_n\}_{n \geq 1} \text{ is absorbed into } \{0\} \mid X_0 = 3) = \frac{1}{40}$
ALTERNATIVE: $\mathbb{P}(\{X_n\}_{n \geq 1} \text{ is absorbed into } \{4\} \mid X_0 = 3) = \frac{39}{40}$ |
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Note:

$$(I_3 - Z)^{-1}B = \begin{pmatrix} \frac{13}{40} & \frac{27}{40} \\ \frac{1}{10} & \frac{9}{10} \\ \frac{1}{40} & \frac{39}{40} \end{pmatrix}.$$

- (d) (1 mark) The number of positive recurrent states is 2

Note: 0 and 4 are absorbing and hence positive recurrent.

- (e) (1 mark) The number of transient classes is 1

Note: $\{1, 2, 3\}$ is the transient class.

- (f) (2 marks) Write down a stationary distribution for this Markov Chain as a row vector (1, 0, 0, 0, 0)

Note: $\lambda(1, 0, 0, 0, 0) + (1 - \lambda)(0, 0, 0, 0, 1), \lambda \in [0, 1]$ gives stationary distributions.

- (g) (1 mark) Is the above stationary distribution unique? Underline the correct answer. Yes/No.

END SEM QUESTIONS: PART B – HINTS

Question 1. Let $\{X_n\}_{n=0}^\infty$ be a Markov Chain with a countably infinite state space S and with the transition probability matrix $P = (p_{ij})_{i,j \in S}$. Assume that $\lim_{n \rightarrow \infty} p_{ij}^n = v_j$ for all $i, j \in S$, where v_j depends on j and not on i . Further assume that the row vector $v = (v_j)_{j \in S}$ satisfies $vP = v$.

Question 1 (a): (2 marks) Show that $\sum_{j \in S} v_j \leq 1$.

Hints: Without loss of generality, denote the states as $0, 1, 2, \dots$. Then, for a fixed i

$$\sum_{j=0}^m v_j = \sum_{j=0}^m \lim_{n \rightarrow \infty} p_{ij}^n = \lim_{n \rightarrow \infty} \underbrace{\sum_{j=0}^m p_{ij}^n}_{\leq 1} \leq 1.$$

Letting $m \rightarrow \infty$ the result follows.

Question 1 (b): (2 marks) Show that $vP^n = v$ for all $n = 2, 3, \dots$.

Hints: The result is true for $n = 1$. We apply the principle of Mathematical induction. Assume that the result is true for $n = 1, 2, \dots, k$. Then,

$$vP^{k+1} = vP^k P = vP = v.$$

Question 1 (c): (2 marks) Show that either $v_j = 0$ for all $j \in S$ or $\sum_{j \in S} v_j = 1$.

Hints: Start with $v_j = (vP^n)_j = \sum_i v_i p_{ij}^n$. Justify taking limits as $n \rightarrow \infty$ by the Dominated Convergence Theorem and show $v_j = v_j \sum_i v_i$. The result follows.

Question 1 (d): (2 + 2 marks) If $v_j = 0$ for all $j \in S$, show that there is no stationary distribution. If $\sum_{j \in S} v_j = 1$, show that $v = (v_j)_{j \in S}$ is the unique stationary distribution.

Hints: If $\pi = (\pi_j)_{j \in S}$ is a stationary distribution, then $\pi_j = (\pi P^n)_j = \sum_i \pi_i p_{ij}^n$. Justify taking limits as $n \rightarrow \infty$ by the Dominated Convergence Theorem and show $\pi_j = (\sum_i \pi_i) v_j = v_j$.

Consequently, if $v_j = 0$ for all $j \in S$, show that there is no stationary distribution. If $\sum_{j \in S} v_j = 1$, then

for any arbitrary stationary distribution π , we get $\pi = v$ and hence $v = (v_j)_{j \in S}$ is the unique stationary distribution.

Question 2. Fix $q \in (0, \frac{1}{2})$ and set $p := 1 - q$. Consider a Markov Chain with the state space $\{0, 1, \dots\}$ and transition probability matrix

$$P = \begin{pmatrix} p & q & 0 & 0 & 0 & \cdots \\ p & 0 & q & 0 & 0 & \cdots \\ 0 & p & 0 & q & 0 & \cdots \\ 0 & 0 & p & 0 & q & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}$$

Question 2 (a): (3 marks) Show that the Markov Chain is irreducible.

Hints: From the transition probability matrix,

$$p_{ii+1} > 0 \quad \& \quad p_{i+1,i} > 0, \forall i \in \{0, 1, \dots\}.$$

Hence, $i \leftrightarrow i + 1$ for all $i \in \{0, 1, \dots\}$ and by transitivity $i \leftrightarrow j$ for all $i, j \in \{0, 1, \dots\}$.

Question 2 (b): (2 marks) Show that the Markov Chain is aperiodic.

Hints: Since there is exactly one communication class and periodicity is a class property, it is enough to find the period of one state.

Now, $p_{00} > 0$ and hence

$$d(0) = g.c.d.\{n > 0 \mid p_{00}^n > 0\} = g.c.d.\{1, \dots\} = 1.$$

Question 2 (c): (5 marks) Is the Markov Chain positive recurrent?

Hints: Show that the system

$$\pi = \pi P, \sum_{i \in S} \pi_i = 1, \pi_i \geq 0, \forall i \in S$$

has a unique solution (for example, using difference equation technique discussed in class). In an aperiodic irreducible chain, if the above system of equations has a unique solution, then the chain must be positive recurrent (this result was discussed in class).